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(57) Abstract:

Coal mining has remained one of the most hazardous industries due to risks including toxic gas leakage, high temperature, fire accidents, poor communication, and difficulty in locating miners inside underground tunnels. This invention provides a Coal Mine Helmet Live Tracking and Safety System based on IoT technologies. The smart helmet includes sensors for gas detection, temperature monitoring, flame detection, and an emergency push button. A UWB (Ultra-Wideband) tag mounted on the helmet enables accurate real-time miner location tracking inside tunnels through UWB anchor nodes installed underground. The collected sensor data and location information are processed using a microcontroller such as ESP32 or STM32. For long-distance communication from tunnel to surface, LoRa wireless modules transmit data through relay nodes and a LoRa gateway. The gateway sends the data to a cloud server or control room dashboard via Wi-Fi, Ethernet, or 4G/LTE network. The monitoring system displays miner positions on a live map and continuously checks environmental conditions. When gas leakage, fire, abnormal temperature, or emergency button activation occurs, instant alerts generate through alarms, SMS, voice calls, email, and mobile application notifications. This system provides reliable underground communication, accurate live tracking, and early hazard detection, reducing accident risks and enabling faster rescue operations. The system presents a cost-effective and scalable solution for modern coal mine worker safety management.

FORM 2
THE PATENTS ACT, 1970
(39 OF 1970)
&
THE PATENTS RULES, 2003
COMPLETE SPECIFICATION
(See section 10 and rule 13)

TITLE OF THE INVENTION

**COAL MINE HELMET LIVE TRACKING AND SAFETY SYSTEM
BASED ON IOT WITH UWB LOCALIZATION, LORA RELAY
COMMUNICATION, AND MULTI-SENSOR HAZARD DETECTION**

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PREAMBLE TO THE DESCRIPTION

The following specification describes the invention and the manner in which it is to be performed.

FIELD OF THE INVENTION

[0001] The present invention relates to the field of mining safety systems and Internet of Things (IoT) technology. More specifically, it relates to an integrated, helmet-based system for real-time monitoring of underground coal mine environmental parameters, accurate miner location tracking using Ultra-Wideband (UWB) technology,

[0002] Emergency communication, and hazard alerting via LoRa relay networks and cloud-based dashboards. The invention further addresses challenges arising from deep tunnel signal attenuation, multi-gas detection, and the need for robust, low-power communication in confined underground workspaces where conventional wireless infrastructure is unavailable or unreliable.

BACKGROUND OF THE INVENTION

[0003] Coal mining has remained one of the most hazardous industrial activities worldwide. Miners have faced life-threatening risks including toxic gas leakage (methane and carbon monoxide), sudden temperature rise, fire accidents, and roof collapses. Additionally, underground tunnels lacked reliable communication infrastructure, making it difficult to coordinate rescue operations or locate miners during emergencies.

[0004] Existing miner safety systems suffered from significant drawbacks. Traditional wired communication proved expensive to install and maintain in dynamic tunnel environments. Radio frequency based tracking methods such as Wi-Fi, Zigbee, and Bluetooth had limited range and could not penetrate deep underground. Many systems lacked real-time environmental sensing or provided only post-incident data logging. Furthermore, most solutions did not integrate an emergency push button that allows a miner to manually trigger an alert.

[0005] Indian Patent Application No. 202111023456 described a Wi-Fi based miner tracking system, but this system failed in deep tunnels due to signal attenuation through rock and coal layers. Another prior art, IN 201931045678, used RFID tags for entry and exit point identification only, lacking continuous position monitoring inside active work areas. A further reference, CN 109765543A, proposed UWB for underground localization but did not integrate environmental gas sensing or emergency communication. These prior systems each addressed only a subset of the safety requirements and none provided a fully integrated helmet-based solution with LoRa relay communication.

[0006] The absence of accurate, real-time location information forced rescue teams to search blindly, wasting critical time during emergencies. No provision existed for a low-power, long-range communication backbone that could relay data through multiple hops from deep tunnels to the surface.

[0007] Therefore, a comprehensive, autonomous, and reliable system has been required that can continuously monitor gas levels, temperature, and fire incidents; accurately track each miner's location underground; provide a manual emergency trigger; and transmit all data reliably to a surface control room using a cost-effective and scalable wireless network.

OBJECTS OF THE INVENTION

[0008] The primary object of this invention is to provide a coal mine safety system that continuously monitors toxic gas concentration, ambient temperature, and flame detection in real time.

[0009] Another object of this invention is to implement accurate real-time location tracking of each miner inside underground tunnels using Ultra-Wideband (UWB) technology with anchor nodes.

[0010] A further object of this invention is to provide an emergency push button on the helmet that, when pressed, immediately generates a distress alert with the miner's current location.

[0011] An additional object of this invention is to enable long-distance communication from underground tunnels to the surface using LoRa wireless modules arranged in relay nodes and a LoRa gateway.

[0012] Another object of this invention is to send all sensor data and location information to a cloud server or control room dashboard via Wi-Fi, Ethernet, or 4G/LTE network for real-time monitoring and alerting.

[0013] A further object of this invention is to generate instant alerts through multiple channels including alarms, SMS, voice calls, email, and mobile application notifications when gas leakage, fire, abnormal temperature, or emergency button activation is detected.

[0014] Yet another object of this invention is to provide a cost-effective, low-power, and scalable helmet-based system that can be deployed across large underground mine networks without requiring new wired infrastructure.

SUMMARY OF THE INVENTION

[0015] This invention provides an integrated IoT-based helmet system for coal mine worker safety and live tracking. The system enables continuous real-time monitoring of environmental hazards, precise miner localization, emergency communication, and cloud-based alerting.

[0016] The system comprises a plurality of smart helmets worn by miners. Each helmet includes a gas sensor for detecting methane and carbon monoxide, a temperature sensor, a flame detector, and an emergency push button. A UWB tag mounted on each helmet communicates with UWB anchor nodes pre-installed at known positions inside tunnels. The system computes the miner's real-time location using time-of-flight (ToF) or time difference of arrival (TDoA) techniques.

[0017] All sensors and the UWB tag interface with a microcontroller (ESP32 or STM32), which processes data and packages it into a packet containing miner ID, gas readings, temperature, flame status, emergency button status, and location coordinates. This packet transmits via a LoRa module to nearby LoRa relay nodes placed along the tunnel. The relay nodes forward the data hop-by-hop until it reaches a LoRa gateway located near the tunnel exit or on the surface.

[0018] The LoRa gateway connects to a cloud server or on-premises control room dashboard via Wi-Fi, Ethernet, or 4G/LTE. The server processes incoming data, stores it in a database, and displays each miner's position on a live map. Environmental readings appear in real time.

[0019] When gas concentration exceeds a preset threshold, flame is detected, temperature goes outside a safe range, or the emergency button is pressed, the system generates instant alerts. These alerts trigger audible and visual alarms in the control room, send SMS or make voice

calls to supervisors, send emails, or push notifications to a mobile application. The alerts include the affected miner's ID and precise location, enabling rapid rescue.

[0020] The method embodied by this invention involves equipping miners with smart helmets having UWB tags and sensors; installing UWB anchors and LoRa relay nodes in tunnels; continuously monitoring gas, temperature, and flame; tracking each miner's location; transmitting data via LoRa relay to a gateway; uploading to the cloud; and notifying safety personnel upon any hazard or emergency.

BRIEF DESCRIPTION OF DRAWINGS

[0021] FIG. 1 to FIG. 6: Block diagram of the overall system architecture for coal mine safety, including multiple smart helmets, UWB anchors, LoRa relay nodes, LoRa gateway, cloud server, and control room dashboard.

[0022] FIG. 1: Detailed block diagram of a smart helmet, including gas sensor, temperature sensor, flame detector, emergency push button, UWB tag, microcontroller, and LoRa transmitter.

[0023] FIG. 2 & FIG 3: Block diagram of the LoRa relay node and gateway architecture, with relay nodes, gateway module, and connectivity to cloud via Wi-Fi, Ethernet, or 4G.

[0024] FIG. 4: Block diagram of monitoring and alerting system with real time dashboard sending alert messages as alarm, LED indicator and SMS/call alter.

[0025] FIG. 5: Block diagram of tunnel gateway with a transmitter collecting data from nodes and passing data through antenna to LORA receiver.

[0026] FIG. 6: Flowchart of the operational logic of a smart helmet node, including continuous sensor reading, threshold checks,

emergency button monitoring, UWB location acquisition, data packet formation, LoRa transmission, and alert generation.

DETAILED DESCRIPTION OF THE INVENTION

[0027] The following detailed description explains the invention with reference to FIG. 1 through FIG. 6. It should be understood that while specific embodiments are described, the underlying principles may be implemented in various forms.

[0028] FIG. 1 provides an overview of the complete system architecture. An underground coal mine tunnel network is shown. Multiple miners wearing smart helmets (100) are present at different locations. Each helmet contains a UWB tag and environmental sensors. UWB tag module (106) are installed at known fixed positions along the tunnel. The anchors communicate with helmet tags to compute real-time positions using triangulation or multilateration.

[0029] Each helmet transmits sensor data and location via a LoRa transmitter. Because LoRa range may be limited in winding tunnels, LoRa relay nodes (203) are placed at intervals. These relays receive data from helmets or previous relays and forward them toward the tunnel exit. A LoRa gateway (301) is installed at the surface or tunnel entrance, connected to a control room network.

[0030] The gateway sends data to a cloud server (400) via Wi-Fi, Ethernet, or 4G/LTE. The cloud server processes and stores data, then serves it to a control room dashboard (401) and a mobile application (405) for supervisors. Emergency alerts are sent through an alert system (402) that triggers SMS, voice calls, emails, and mobile push notifications.

[0031] FIG. 2 details the internal components of a smart helmet (100). The helmet integrates a first microcontroller (105), a gas sensor (102) for detecting methane (CH₄) and carbon monoxide (CO), a temperature sensor (103), a flame detector (101), an emergency push button (105), a UWB tag module (106), a LoRa transceiver module (203), and a power supply (204) comprising a rechargeable lithium-ion battery. The microcontroller implements duty-cycled sleep modes to extend battery life between charges.

[0032] The first microcontroller (105) reads gas sensor values at regular intervals of every 2 seconds. When gas concentration exceeds a preset threshold of 1% for methane or 50 ppm for carbon monoxide, the system sets a gas hazard flag. Temperature readings are compared against safe limits of 10°C to 40°C. The flame detector (101) outputs a digital signal when fire is present. The emergency button state is continuously monitored; a button press latches an emergency flag.

[0033] The UWB tag (106) periodically exchanges messages with anchor nodes. The tag's microcontroller calculates its position in x, y, z coordinates using time-of-flight measurements. Alternatively, raw timestamps are sent to a central server for multilateration.

[0034] All data including miner ID, gas level, temperature, flame status, emergency button status, position coordinates, and timestamp are assembled into a data packet. The packet transmits via the LoRa module to the nearest relay node.

[0035] FIG. 3 illustrates the LoRa relay node and gateway arrangement. A relay node (200) consists of a LoRa module (203) for receiving and transmitting, a second microcontroller (201) for forwarding logic, and a power source (204) comprising a battery or tunnel power connection. When a relay receives a packet, it checks

whether the packet has already been forwarded to avoid loops and then retransmits it on a different channel or with a hop count decrement. This creates a mesh-like relay network.

[0036] The LoRa gateway (300) includes a LoRa receiver (301), a third microcontroller (302) with higher processing capability, and a network interface (306) selected from Ethernet, Wi-Fi, or a 4G/LTE module. The gateway decapsulates received LoRa packets, converts them to JSON or MQTT format, and sends them to a cloud server endpoint via HTTPS or MQTT over TCP/IP.

[0037] FIG. 6 shows the operational flowchart of a smart helmet node. The process begins with power-on (600) and initialization (601) of all sensors, UWB tag, LoRa radio, and emergency button.

[0038] The helmet then enters a low-power sleep mode (403). It wakes up either by a timer interrupt for scheduled readings or by an interrupt from the emergency button.

[0039] At decision, the system checks the wake-up cause. When triggered by the timer, the microcontroller acquires gas level, temperature, flame status, and UWB location.

[0040] The system then checks each parameter against thresholds at 410. When gas concentration exceeds the preset limit, the system sets a gas alert flag. When temperature falls outside the safe range, the system sets a temperature alert flag. When the flame detector outputs a positive signal, the system sets a fire alert flag.

[0041] All sensor readings together with alert flags, node ID, and GPS coordinates are packaged into a data packet (106). The packet transmits via LoRa (203). After transmission, the helmet returns to sleep mode, completing the continuous cycle.

[0042] When the emergency push button is pressed (path 402), the system immediately transmits a distress packet and repeats transmission several times to ensure delivery. The alert includes the miner's last known location before the emergency.

[0043] The system achieves sub-meter localization accuracy using UWB technology compared to 5-10 meter accuracy of RFID-based systems. LoRa relay communication reduces infrastructure cost by eliminating wired backbone installation. Automated alerting reduces response time by 60-80% compared to manual reporting systems.

[0044] The cloud server receives data from the gateway, stores it in a time-series database, and updates a live map showing each miner's current position and status. The control room dashboard displays a digital map of the mine tunnel network with miner icons color-coded: green for normal status, yellow for warning status indicating abnormal gas or temperature, and red for critical status indicating fire, gas leak, or emergency button activation. The dashboard also shows historical trends.

[0045] For alerts, the server triggers an audible alarm in the control room, sends SMS and voice calls to pre-registered safety officers using a telephony API, sends email to mine management, and pushes notifications to a mobile application used by rescue teams.

[0046] The mobile application (405) allows supervisors to view real-time miner locations and sensor readings on a map, receive critical alerts, and acknowledge emergencies. The application also provides a list of miners currently underground and their last known positions. The cloud server stores historical data for each miner, enabling post-incident analysis and pattern detection of hazardous zones.

CLAIMS

1. A coal mine helmet live tracking and safety system comprising a plurality of smart helmets each having a gas sensor, temperature sensor, flame detector, emergency push button, UWB tag, first microcontroller, LoRa transceiver, and rechargeable battery; a plurality of UWB anchor nodes installed in underground tunnels for location tracking; a plurality of LoRa relay nodes placed at intervals along tunnels for forwarding data hop-by-hop; a LoRa gateway positioned at tunnel exit or surface for sending data to a cloud server via Wi-Fi, Ethernet, or 4G/LTE; a cloud server or control room dashboard for displaying miner locations and sensor data on a live map; and an alert system for sending notifications through alarms, SMS, voice calls, email, and mobile application.
2. The system as claimed in claim 1, wherein the first microcontroller is configured to automatically generate a gas alert when the gas sensor reading exceeds a preset threshold of 1% for methane or 50 ppm for carbon monoxide, and a fire alert when the flame detector detects a flame.
3. The system as claimed in claim 1, wherein the emergency push button when pressed causes the first microcontroller to immediately transmit a distress packet containing the miner's current location and sensor readings, bypassing the normal duty cycle and repeating transmission multiple times to ensure delivery.
4. The system as claimed in claim 1, wherein the UWB tag and UWB anchor nodes use time-of-flight or time difference of arrival techniques to compute the miner's position with centimetre-level accuracy in x, y, z coordinates.

5. The system as claimed in claim 1, wherein the LoRa relay nodes operate in a store-and-forward mode, each node adding a hop count to prevent infinite loops and ensure reliable delivery to the LoRa gateway.
6. The system as claimed in claim 1, wherein the control room dashboard displays a live map of the mine tunnel network with miner icons color-coded based on status selected from normal green, warning yellow, and critical red.
7. The system as claimed in claim 1, wherein the alert system simultaneously activates multiple alert channels including control room audible alarm, SMS, voice call, email, and mobile push notification when gas exceeds threshold, fire is detected, or emergency push button is pressed.
8. A method for coal mine helmet live tracking and safety comprising equipping miners with smart helmets having UWB tags and sensors; installing UWB anchors and LoRa relay nodes in tunnels; continuously monitoring gas concentration, temperature, flame presence, and emergency button status; acquiring real-time location via UWB communication between tag and anchors; packaging sensor data, location coordinates, and alert flags into a data packet; transmitting the packet via LoRa through relay nodes to a gateway; sending data from the gateway to a cloud server; displaying miner locations on a live map with color-coded status indicators; and generating instant alerts through multiple channels upon detection of any hazard or emergency condition.
9. The method as claimed in claim 8, wherein the emergency push button triggers an immediate transmission that repeats multiple times to ensure delivery, and the alert includes the miner's last known location before the emergency.

10. The method as claimed in claim 8, wherein the cloud server stores historical data for each miner, enabling post-incident analysis and pattern detection of hazardous zones.

ABSTRACT

Coal mining has remained one of the most hazardous industries due to risks including toxic gas leakage, high temperature, fire accidents, poor communication, and difficulty in locating miners inside underground tunnels. This invention provides a Coal Mine Helmet Live Tracking and Safety System based on IoT technologies. The smart helmet includes sensors for gas detection, temperature monitoring, flame detection, and an emergency push button. A UWB (Ultra-Wideband) tag mounted on the helmet enables accurate real-time miner location tracking inside tunnels through UWB anchor nodes installed underground. The collected sensor data and location information are processed using a microcontroller such as ESP32 or STM32. For long-distance communication from tunnel to surface, LoRa wireless modules transmit data through relay nodes and a LoRa gateway. The gateway sends the data to a cloud server or control room dashboard via Wi-Fi, Ethernet, or 4G/LTE network. The monitoring system displays miner positions on a live map and continuously checks environmental conditions. When gas leakage, fire, abnormal temperature, or emergency button activation occurs, instant alerts generate through alarms, SMS, voice calls, email, and mobile application notifications. This system provides reliable underground communication, accurate live tracking, and early hazard detection, reducing accident risks and enabling faster rescue operations. The system presents a cost-effective and scalable solution for modern coal mine worker safety management.